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A data-guided analytical framework for assessing the value of additional variables in heart transplant decision making

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ABSTRACT

Heart transplantation is a life-saving procedure for patients with end-stage heart failure. The United Network for Organ Sharing (UNOS), which administers the US organ allocation system, substantially expanded the number of clinical and demographic variables collected in its database in 2004. This study examines whether these newly added variables improve the ability to predict survival outcomes for patients on the heart transplant waiting list. An information-gain-based feature selection approach, supported by an extensive review of prior studies, was combined with survival analysis and regularized regression to identify the most influential predictors. Using the selected variables, several classification models, including tree-augmented Naïve Bayes, logistic regression, support vector machines, decision trees, and random forests, were developed. Class imbalance was addressed through random under-sampling and cost-sensitive modeling. The results show that prediction accuracy for short-term, medium-term, and long-term survival (one month, one year, and five years) does not improve substantially when the new variables are included. The findings suggest that the expanded data collection introduced in 2004 adds limited incremental value for predicting survival among patients awaiting heart transplantation.

1. Introduction

Heart failure is a prevalent global health condition affecting approximately 2%–3% of the adult population, or over 26 million people [1]. While the term may suggest the heart has ceased functioning, it actually describes a clinical state where the heart is too weak to pump an adequate volume of blood to meet metabolic demands [2]. In the United States alone, Heart failure affects an estimated 5.8 million individuals, with more than 550,000 new cases diagnosed annually [3,4].

Heart transplantation procedures are typically required for patients having severe end-stage heart failure (meaning all possible treatments except transplant have failed). A careful process is used to select patients in this category for heart transplant procedures. If a patient is eligible, then the patient is placed on a waiting list for a transplant until a suitable donor heart is found [3]. Recently, there has been a dramatic increase in the demand for heart transplantation due to reasons such as aging of the population, increasing obesity, diabetes [5] and an increasing life expectancy [6]. Unfortunately, there has been a huge gap between supply donated and healthy graft, and demand, leading to longer waiting times and patients dying while on the waiting lists [6]. On average, there are approximately 3000 patients waiting for a heart transplant while the annual supply is around only 2000 [3].

In order to optimize the utilization of donated hearts, it is vital to have adequate information to match donated hearts with patients having the best chance of survival after receiving a donated heart. As such, having a reliable data source for use in making donor–recipient matching decisions is crucial. In the US, the United Network for Organ Sharing (UNOS), a private, non-profit organization under contract with the federal government, is tasked with managing the organ transplant system [4]. This system contains datasets regarding heart, lung, liver, kidney, pancreas and intestine organ donations, as well as, transplant

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events occurring in the U.S. since October 1, 1987. These datasets include clinical and demographical information about both donors and recipients, as well as, the transplant procedures itself. It is safe to say the datasets gathered has become a common source that can serve the interests of researchers and the public.

The ability to accurately predict the compatibility between a donated organ and a recipient patient is very important to optimize benefits from the use of donated organ. Having the right data, containing key variables, that are helpful for prediction purposes is essential. As a result, this presents a strong incentive to gather as much data from donors and patients as possible. However, gathering more variable though essential, may not necessarily translate to improved prediction outcomes.

This study investigates the predictive impact of variables added to the UNOS database in 2004 on post-heart transplant survival. Rather than focusing on donor–recipient matching, this research prioritizes post-transplant survivability—a critical factor in clinical decision-making. By accurately categorizing a patient's survival likelihood, practitioners can make informed decisions regarding the frequency of follow-up visits and the intensity of monitoring for complications such as blood clots, strokes, and Coronary Allograft Vasculopathy (CAV). Enhanced risk-stratification is essential for optimizing post-transplant care and ensuring the long-term recovery of high-risk patients.

In this study, our primary objective is to evaluate whether newly introduced clinical and registry-level variables provide incremental predictive value when added to an established feature set for post-transplant survival prediction. Rather than assessing the standalone predictive power of these variables in isolation, we focus on their marginal contribution within an existing modeling framework that already incorporates well-validated predictors. This design reflects real-world clinical modeling practice, where new variables are typically considered as additive enhancements to existing risk models. By explicitly testing whether such additions yield meaningful performance gains, our work directly addresses an important and underexplored question in transplant analytics: whether expanding feature sets necessarily translates into improved predictive performance.

The remainder of this paper is organized as follows. Section 2 provides a literature review of the relevant history for this study. In Section 3, the overall data analytics methodology used in this study as well as the data source and preparation are described. The results obtained from the actual data analysis are provided and discussed in Section 4. In Section 5, conclusions are summarized and some thoughts on the direction of future research are offered.

2. Literature review

In the United States, donor hearts allocation is performed with the objective of minimizing heart transplant waitlist mortality and donor organ wastage. The chances of achieving this objective can be enhanced by properly identifying risk factors for disparities in survival. A tiered system that uses 6 active tiers to reflect the medical urgency of candidates on the waiting list is used to achieve this objective. Efforts have been made to investigate variable in the UNOS database to identify variables that are important in predicting mortality. Variable that were shown to be weak predictors of waitlist mortality included history of stroke, blood type, race, history of tobacco use, vascular disease, prior malignancy, while variables such as estimated glomerular filtration rate, extracorporeal membrane oxygenation, renal function, peak oxygen capacity, age, use of devices like ventricular assist device and mechanical ventilation [7,8]. Although the ability to collect more data has been seen as a necessary requirement in order to improve the ability to adequately predict heart transplant outcomes, excessive data collection may lead to over burdening transplant centers. In addition, study results suggest that not all data collected are helpful in predicting transplant outcomes [9]. However, the drive to collect additional data for prediction purposes is strong.

Over time, UNOS has made significant changes in the number of variables used by increasing the number of variables gathered in its Organ Transplantation Databases. This can potentially can add more information to the existing body of knowledge. For example, UNOS added 165, 85 and 105 new variables to the thoracic transplant database in 1994, 1998 and 2004, respectively.

Unfortunately, data collection process in any field of study is a time consuming and expensive process. To reduce the burden of collecting data manually, automated data collection mechanisms have been widely adopted, thereby making such devices to become more popular and less expensive. However, manual data collection still plays a crucial role in healthcare settings where activities such as blood sample collection, laboratory tests and medical imaging are performed manually.

There are numerous "*benefits*" of collecting more data, in organ transplant setting. Such benefits include enhanced efficiency of donor–recipient matching optimization, sequential medical decision-making processes and predicting the survival life of the recipients (so that aggressive interventions can be done for the high risk-level recipients), etc.

Apart from the cost associated with data collection and the extra burden on patients, other issues that could impact survival life (SL) and quality of life (QOL) may also arise. In healthcare setting, survival life refers to a patient's life expectancy after heart transplantation process is performed, while quality of life (QOL) is a term used to describe a patient's quality of life relative to the patient's health or disease status.

Collecting more data (i.e. more features) and therefore more information (per patient) might bring overfitting issues and expensive computation times in *survival life (SL)* and *quality of life (QOL)* prediction models, in addition to the suboptimal decisions in matching procedures etc. Thus, all the benefit aspects should be carefully analyzed before making such investment, especially in situations where the healthcare organizations decide to aggressively increase the amount of data that will be collected.

Effective utilization of healthcare data can improve the potential of saving many lives. More specifically, improving the prediction of survival is a critical task, because it is an essential tool in donor/recipient matching procedure. Such improvement could in turn lead to a decrease in the gap between supply and demand. Recent research has shown that the information exists in these large/complex UNOS datasets can be analyzed effectively in extracting hidden, novel patterns by employing data mining methods [2–4,10–12]. Compared to traditional statistical techniques, data mining approaches provide a relatively faster means of finding complex and nonlinear patterns from a large amount of data containing many predictors [13,14]. In addition, these methods do not require prior knowledge about the data, nor do they make assumptions about the statistical distribution or properties of the data [15].

Extensive work has been done on survival prediction using data-driven approaches in organ transplantations. Studies have also shown that high prediction outcomes can be achieved when multiple variables are used in the studies [16]. However, this may depend on the actual variables used as some variables may not necessarily be strong predictors of survival outcomes. For instance, studies have shown contradicting effect of gender on the survival of patients after undergoing heart transplantation procedures. Based on their study, Tsao et al. [17] concluded that the effect of the donor or recipient gender, or donor–recipient gender combination on patients' survival was not substantial. However, Eifert et al. [18] surmised that female donors and male recipients' combinations had a lower survival after a year, while male donors and female recipients' combinations resulted in favorable short-term outcomes.

Extensive work has been done on survival prediction using data-driven approaches in organ transplantations. Existing literature on the subject matter can be categorized into three main categories. The first category stream consists of studies that utilized conventional statistical methods (e.g. survival analysis and regression models) to predict a patient's survival after a transplant procedure and also to identify the most important predictors for survival. This category can be categorized further into three subcategories as follows; (a) studies examining the effect of a single variable on survival after a transplant [19,20], (b) studies investigating the effect of more than one variable by employing conventional statistical approaches [21–24] and, finally (c) studies creating risk scores or levels by using the effect of multiple variables on the outcome after transplants [3].

For studies under the second category, machine learning-based analytical methods employed to predict survival. The main difference between the first and the second categories can be seen as a philosophical difference between both streams [25]. Statistical methods focus on the efficient collection of data for the purpose of answering specific questions. On the other hand, machine learning based approaches seek to find unsuspected relationships and patterns from collected data. As an example, Kusiak et al. [10] compared the results of two rule-based data mining techniques, decision trees and rough sets, for predicting the survival time of kidney dialysis patients. Decision trees (DT) were also used by Nakayama et al. [26] to predict the prognosis of acute liver failure in an effort to improve the indication criteria for liver transplants. Kaplan and Schold [27] compared the power of an artificial neural network (ANN) and a statistically derived nomogram in predicting the 5-year graft survival after kidney transplants based on demographic, clinical and pharmaceutical data. Oztekin et al. [15] compared the predictive performance of ANN, logistic regression, support vector machines (SVM) and decision tree for long-term survival after a thoracic transplantation. Recently, Dag et al. [3] employed a probabilistic data mining approach (i.e. Tree-augmented Naïve Bayes (TAN) algorithm) to obtain patient-specific survival likelihood score for the patients who underwent heart transplant, and to investigate the conditional interdependencies between the predictors. In general, it can be concluded that all the aforementioned data-driven studies have uncovered useful hidden patterns and information embedded in the large transplantation datasets. Ayers et al. [28] evaluated machine-learning models for predicting survival after heart transplantation using UNOS registry data. They compared several algorithms (including random forests and ensemble methods) with traditional Cox regression and found that machine-learning models achieved higher predictive performance. Their study showed that ML approaches better captured complex relationships among donor and recipient variables, improving risk stratification and clinical prediction accuracy. Lisboa et al. [29] developed explainable artificial intelligence (XAI) models to predict survival after heart transplantation. Using registry data, they showed that machine-learning models achieved strong predictive performance while also providing model interpretability through explainability techniques. Their study identified key clinical predictors and demonstrated how XAI can support transparent and clinically meaningful decision-making in transplant risk assessment. Recently, Liou et al. [30] evaluated contemporary machine-learning models for predicting post-heart transplant mortality using recent-era clinical data. Their study compared multiple ML algorithms and demonstrated improved predictive performance over traditional statistical approaches, highlighting the value of modern ML techniques in capturing complex risk patterns in current transplant populations. The authors emphasized model robustness and applicability to present-day clinical practice.

In alignment with these machine learning-based analytical approaches, prior research has applied predictive analytics and machine learning to healthcare decision problems, underscoring the relevance of our study. Kuvvetli et al. [31] developed an artificial neural-network-based predictive analytics model for COVID-19 case and mortality forecasting, demonstrating how data-driven models can support high-stakes public health and resource-allocation decisions. Sawhney et al. [32] conducted a comparative assessment of artificial intelligence models for early prediction and evaluation of chronic kidney disease, emphasizing model selection, feature representation, and performance evaluation in a clinical setting. More recently, Khanna et al. [33] proposed a machine learning and explainable artificial intelligence triage-prediction system for COVID-19 severity, combining feature selection, class-imbalance handling, and interpretable models to support ICU triage.

Finally, the third category investigated the dynamic effect of significant variables on the survival outcome. Most of the work within this stream category is based on statistical techniques [34–36], with the exception of the study performed by Lin et al. [11] who determined the effect of the predictors on 1-year to 7-year survival by using logistic regression, Cox Proportional Hazard Model (CPHM), and multiple-output ANNS. Another exception, Hsich et al. [9] analyzed the Scientific Registry of Transplant Recipients (SRTR) database to identify the most important variables predicting waitlist mortality among heart transplant candidates. Using advanced statistical and machine-learning methods, they ranked key clinical, demographic, and hemodynamic factors associated with mortality risk while patients were awaiting transplantation. Their findings highlighted the critical role of disease severity indicators, organ function markers, and clinical status variables in risk stratification. This study provides an important foundation for understanding pre-transplant mortality risk and informs feature selection for predictive modeling in heart transplantation.

Based on existing literature, studies have employed data analytical approaches (both machine learning and conventional statistical methods) to predict the survival outcome. While prior studies have explored a wide range of machine learning approaches for transplant outcome prediction, most implicitly assume that incorporating additional variables will improve model performance. However, systematically evaluating the incremental benefit of newly proposed predictors when added to established clinical feature sets seems to be a tangible gap. This gap is particularly relevant in large registry-based studies, where increasing model complexity may introduce noise and reduce interpretability without delivering tangible performance gains. Our work addresses this gap by formally testing this assumption across multiple algorithms and time horizons, thereby contributing empirical evidence to an area that has largely been guided by intuition rather than rigorous validation.

The main goal of this study is to evaluate the contribution of the newly added variables in 2004 in predicting the survival outcome after heart transplants. In other words, our goal is to examine the "*benefits*" of collecting more data from the prediction of survival life (short-, mid-, and long-term) perspective only. As mentioned in the earlier sections, in the proposed study, our purpose is not to focus on the donor–recipient matching. But it is rather to focus on the survivability after the transplant, which is still critically important from the clinical decision-making perspective. A healthy information about patient's survival likelihood category, the medical practitioners can give more effective/time decisions on follow-up visit frequencies, and/or may be (less or more) aggressive in monitoring for post-transplant events such as blood clots (which may cause heart attacks), strokes, lung issues/problems, kidney functions, coronary allograft vasculopathy (CAV) etc. Post-transplant care is critically important for a healthy recovery, and some of these events should be taken more seriously if the patient is labeled as a "high risk" one. In the proposed study, a data analytical methodology is employed that aims at investigating; (1) which of these new variables are selected by powerful variable selection algorithms, (2) whether there is any meaningful and consistent improvement in the predictability of the outcomes, when these variables are included in the prediction analysis, and (3) the contribution of these variables in predicting the outcome by applying sensitivity analysis.

3. Methodology

In this study, a comprehensive data analytical methodology that consists of four sequential phases is offered. As depicted in Fig. 1, the first phase includes 3 steps, where in the first step, the transplants that happened before 2004 are discarded from the dataset. The rationale

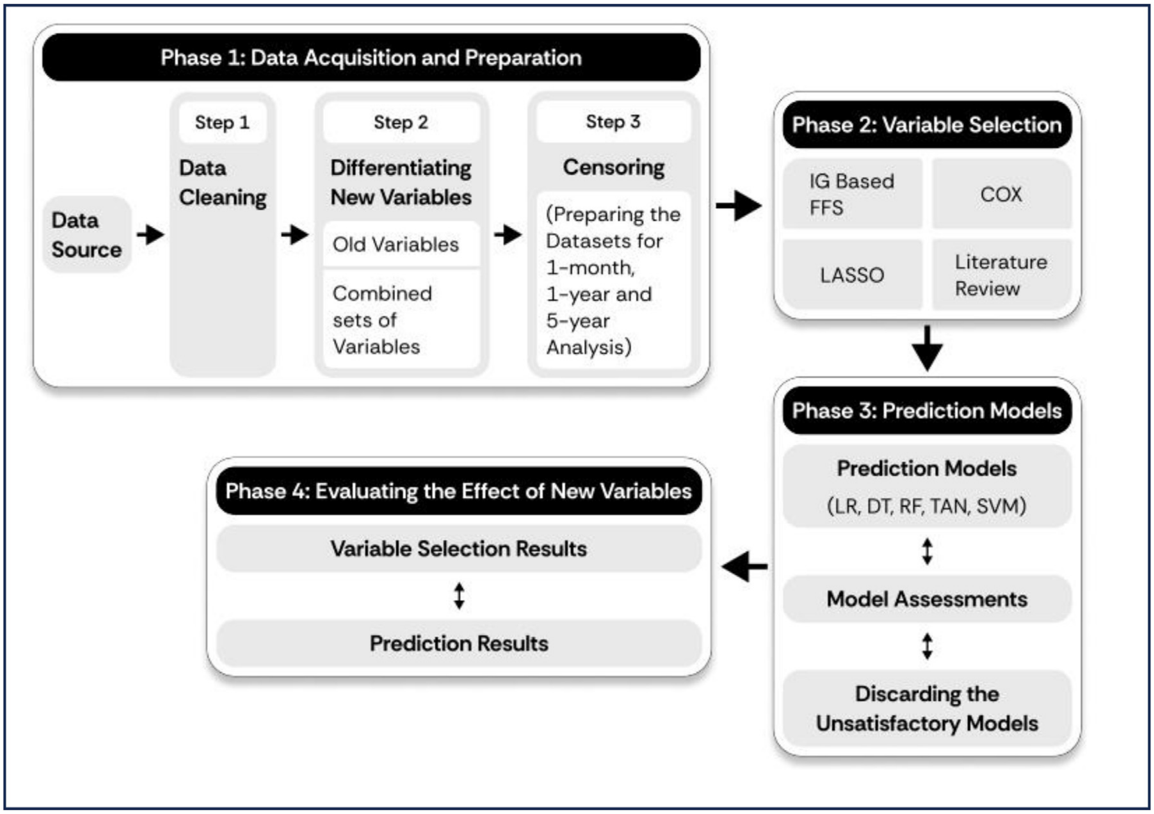

Fig. 1. An overview of the proposed methodology.

behind that is that the overall goal of the proposed study is to measure the effect of the variables that were added to the UNOS dataset in 2004. This leads us to perform two separate analyses; (1) including, and (2) excluding the variables that were added in 2004. (Throughout the paper, we call these group of variables "*combined*" variables and "*old*" variables, respectively, for simplicity). After obtaining the dataset, the dataset was prepared in a systematic way in step 1. In step 2, the entire set of variables are differentiated as; (1) *old* and (2) *combined* variables. In step 3, three different datasets were extracted from each dataset to evaluate the effect of the new variables in different time horizons (1- month, 1-year and 5-years) via censoring. In Phase 2, well-known and suitable feature selection algorithms were applied to select the most important variables to both datasets. Phase 3 represents the process in which different data mining classification algorithms were applied by deploying the most important features that were selected in the previous phase. In addition, the performance of these models was assessed, and the unsatisfactory ones were discarded in the same phase. In the final phase (Phase 4), the contribution of the new variables was measured via sensitivity analysis. This process is repeated for both 1-month, 1- and 5-year analysis. More detailed explanation about the procedure is provided in the following related sections.

3.1. Data acquisition and preparation

The UNOS dataset contains information about the heart transplantation cases that have been performed between October 1, 1987 and December 31, 2012. It includes clinical and demographical variables about both donors and recipients as well as the transplant procedure itself. These variables can be grouped into three main categories such as; (1) pre-operative variables which includes *age, gender, blood type* etc., (2) the surgical procedural variables which includes if there is a *chronic steroid use* at transplant, or if the recipient is on any type of life support at the time of transplant and, (3) post-operative variables such as if the patient has *acute rejection* or *length of hospital stay* after the operation etc. The dataset was de-identified by UNOS prior to being received by our research team.

3.1.1. Data cleaning and differentiating the recently added (new) variables

UNOS has made notably different changes in the number of variables in its Organ Transplantation Databases since October 1, 1987, in order to store more information that might in turn be used by the medical research community. For example, there are 165, 85 and 105 variables, which have been added to the Thoracic Transplant database in 1994, 1998 and 2004, respectively. As discussed earlier, our end goal in this proposed study is to evaluate the effect of these changes (in the number of variables) in predicting the survival outcome after heart transplants. In order to meet such research goal, 2004 has been selected by the research team in order to focus on one of these three different time points (i.e. 1994, 1998 and 2004). The rationale behind this is that the surgical technology that is being used for heart transplant procedures, and thus the post-transplant survival time have changed in the recent years [35]. Under such circumstances, it would be naïve to compare the prediction performances of the classification models on the survival outcomes of the transplant cases that were performed in 1994 with the ones performed in 2012. That would be more objective to compare them on the cases that were performed in a narrower time frame. Therefore, the transplants performed prior to June 30, 2004 were eliminated from the dataset.

In this study, the datasets were prepared in a systematic way such that invariant variables, outliers and the variables that have no effect on the survival outcome such as patient ID, transplant ID, transplant date have been checked for and eliminated. It should be noted that the UNOS dataset has excessive number of missing records, and one should be very careful in dealing with the situation where there are a lot of missing records, which in turn might lead the classification models to suffer. Dealing with missing values, the three common data cleaning approaches namely *Case-wise deletion, Column-wise deletion, and imputation* have their peculiar disadvantages for such situations.

Table 1
18 variables added to UNOS heart transplant databases after 2004 (after cleaning the dataset).

Variable name	Description
ALCOHOL_HEAVY_DON	Heavy Alcohol Use (heavy=2+ drinks/day)
ARGININE_DON	Deceased donor-was donor given arginine vasopressin
BMI_TCR	BMI at listing
CDC_RISK_HIV_DON	Does the deceased donor meet CDC guidelines for high risk for an organ donor
CIG_USE	
HEMATOCRIT_DON	Hematocrit
INSULIN_DON	Deceased donor-was donor given insulin?
INOTROP_VASO_CO_TCR	most recent CO L/MIN INOTROPES/VASODILATORS YES/NO at listing
INOTROP_VASO_DIA_TCR	most recent PA (DIA) MM/HG INOTROPES/VAOSDILATORS YES/NO at listing
INOTROP_VASO_MN_TCR	most recent PA (MEAN) MM/HG INOTROPES/VASODILATORS YES/NO at listing
INOTROP_VASO_PCW_TCR	most recent PCW (MEAN) MM/HG INOTROPES/VASODILATORS YES/NO at listing
PCO2_DON	PCO2
PH_DON	Blood pH
TCR_DUR_ABSTAIN	Duration of abstinence for cigarette use
TRANSFUS_TERM_DON	Number of transfusions during this (terminal) hospitalization:
VAD_DEVICE_TY_TCR	Candidate type of VAD device at listing
VAD_DEVICE_TY_TRR	VAD device type
WORK_INCOME_TCR	Work for income at registration?

Case-wise deletion would cause a substantial decrease in the number of samples, which may adversely affect the ability of the data mining model to train effectively. In the case of *column-wise deletion*, there is a tendency of having few variables left after the cleaning process is completed and may cause a significance information loss in the model. The imputation approach has the advantage of keeping the most information during the cleaning process. In this study, fast imputation methods were used. The missing numerical variables values are imputed with *median* and for categorical variables, the missing observations are labeled as "*UNKNOWN*".

The rational for imputation with *median* is twofold: the first is to achieve a complete case analysis, which may be compromised if a *Case-wise* or *Column-wise* deletion approach is employed [37,38]. The second reason is that imputation with *median* is less susceptible to skewness [39] and very efficient for imputing large datasets. Similarly, labeling the missing observations of categorical observations with "*UNKNOWN*", keeps the maximum observations and is computationally favorable for large datasets such as UNOS transplantation registry.

As noted earlier, the primary objective of this study is to measure the effect of the newly added variables in predicting the survival outcome after the heart transplants. Therefore, four post-operative variables can be employed as candidate survival outcomes of the proposed study. These variables are *gstatus* (a binary variable denoting if the graft has survived or failed at the last follow-up time), *pstatus* (a binary variable denoting if the patient has survived or failed at the last follow-up time), *gtime* (a continuous variable denoting the time range between the transplant date and graft's last follow-up/failure time, in days) and *ptime* (a continuous variable denoting the time range between the transplant date and person's last follow-up/death time, in days). Since our purpose to focus on the patients who solely died due to heart-transplantation related reasons, the *gstatus* variable along with *gtime* are the suitable outcomes for our study. The *gstatus* is selected as our main target, however *gtime* will also be incorporated throughout the study in that it enables us to censor the datasets, which is described in the subsequent section in detail. Therefore, the *pstatus* and *ptime* variables are also eliminated from the datasets. Having applied such cleaning process have left us with **25,512** cases and **161** variables of which **18** are new and **143** are old variables (see Table 1).

3.1.2. Censoring

In this study, three different time points (1-month, 1-year and 5-years) are determined as critical points in evaluating the performances of the models generated using the classification algorithms. There are several reasons for us to choose these three specific time points. One of the primary reasons is that these time points are commonly studied in the existing heart transplant literature [23,36,40–43]. In addition, the reason that we have not included larger time horizons such as 6-, 7-, 8-, 9- or 10-years analysis is because of the limitation on the sample size, since we only include the transplants performed after 2004. Another important reason to study specifically 1-year and 5-year time points is that they are considered as the reference time points in providing the common survival rate statistics after the cardiac transplant [35].

Table 2
Number of survivals, failures, and censored observations over the time-points.

Time point	Survivals	Failures	Censored
1-month	23 722	1165	625
1-year	19 704	2693	3115
5-year	9788	4956	10 768

For each time point that is used in this study (1-month, 1- and 5-year), some of the cases in the dataset are censored with the following procedure. For a case to be censored, the patient should be alive on the last follow-up date and the last follow-up date should be less than the number of days required (1* 365 = 365 days for 1-year analysis) for the analysis. The reason for those patients to be eliminated from the analysis is that; it is not known whether they will be still alive or not, after the required day. In addition, the *gtatus* values of the patients who have died after the number of days required should be changed from "*1*" to "*0*". The reason behind that is, the patient has died sometime after the threshold value. However, from a classification point of view, he/she has succeeded by living longer than the predetermined time point. This censoring procedure can be represented by the following pseudo code:

If $(gstatus\ = 0\, and\ gtime < \#\,of\ days\ required)$ **then** Censor the case
Elseif $(gstatus = 1\ and\ gtime > \#\,of\ days\ required)$ **then** gstatus $= 0$

After applying the censoring procedure, a summary of distribution of survivals, failures and censored cases are represented in Table 2. It should be noted that the sum of survivals, failures and *censored observations* for each row is 25,512 as it was discussed in Section 2.1.1.

3.2. Variable selection

As discussed in Section 3.1.1, after the cleaning procedure, the dataset includes 161 variables in total (i.e., 143 existing and 18 new variables). Deploying hundreds of variables (as in this case) into any prediction model would both computationally be very expensive and carry a high risk of overfitting. In such situations, feature selection methods can be employed in selecting the suitable subset of the existing variables. Existing feature selection methods can broadly be categorized into two groups i.e. filters and wrappers [44,45]. Wrapper methods are specifically dedicated to a specific type of prediction method. They rely on the performance of one type of classifier to evaluate the quality of a set of features. On the other hand, the filter methods can be considered as agnostic models in that they rely on general characteristics of the training data to select some features without involving any learning algorithm. The filter methods rank features according to their individual predictive power, which can be estimated by various means such as Fisher score, *Kolmogorov–Smirnov test*, *Pearson correlation* or *mutual information* [46]. Both approaches have their own advantages, which depend on the number of features, samples and the existence of complex nonlinear relations among the variables. In our analysis, we applied three popular variable selection approach such as *Information gain-based feature selection* (IG-FFS), *Cox Regression*, and *LASSO* to extract the most important features. In addition to the selected variables (by feature selection algorithms), the variables that have been found important in the existing literature have also been added to the final set of predictors. So, we adopted an intentionally inclusive feature screening strategy by combining variables identified through IG-FFS, LASSO, Cox regression, and prior literature. This union-based approach was designed to minimize the risk of prematurely excluding potentially informative predictors. Rather than enforcing artificial consensus across selection algorithms—which are known to capture complementary statistical perspectives—we allowed any variable flagged by at least one principled method to enter the candidate feature pool. Importantly, all feature selection procedures were performed within each training fold of the cross-validation framework, ensuring methodological validity and preventing information leakage. This strategy enables a conservative assessment of whether newly introduced variables can contribute incremental predictive value under generous inclusion criteria.

It should be noted that, to prevent data leakage and ensure unbiased performance estimation, variable selection was fully nested within the cross-validation procedure. Specifically, for each fold of the 5-fold cross-validation, feature selection using IG-FFS, LASSO, and COX was performed exclusively on the training subset of that fold. The selected features were then used to train the corresponding predictive model, which was subsequently evaluated on the held-out test subset. This fold-wise feature selection strategy ensures that no information from the test data is used during the feature selection process. Only variables that were consistently selected across cross-validation folds were considered stable predictors and retained for downstream analysis. This approach aligns with widely accepted best practices in predictive modeling and machine-learning methodology and supports the reproducibility and robustness of the reported results.

3.2.1. Fast feature selection (FFS) via information gain analysis

In prediction models, a feature can generally be considered as "good" if it is relevant to the target but is not redundant to any of the other predictors in the dataset. If the correlation between two variables is adopted as a "goodness" measure, it can be concluded that a feature can be selected if it has a high enough correlation with the target (which makes it "predictive of" the class) and a low enough correlation with any other variables (which is not above a certain threshold level). In this sense, a suitable measure of correlation between features needs to be employed. The existing correlation measures can be mainly classified into two groups; (1) based on classical linear correlation and, (2) based on information theory [47]. However, in the situations where there are hundreds of variables (as in our case), it would be very naïve to assume linear correlation between features. Therefore, in this study, a correlation measure that is based on the information-theoretical concept of entropy is adopted. The entropy of a variable X is defined as

$$H(X) = -\sum_i P(x_i).\log 2(P(x_i)) \tag{1}$$

and the entropy of X after observing values of another variable Y is defined as

$$H(X) = -\sum_i P(x_i).\log 2(P(x_i)) \tag{2}$$

where P(x_i) is the prior probabilities for all values of X, and P ($x_i|y_i$) is the posterior probabilities of X given the values of Y. The amount by which the entropy of X decreases reflects additional information about X provided by Y and is called information gain that is given by

$$IG(X|Y) = H(X) - H(X|Y) \tag{3}$$

According to this measure, a feature Y is regarded more correlated to feature X than to feature Z, if $IG(X|Y) > IG(Z|Y)$. In other words, the features are selected based on the increase in the amount of information (gained) by knowing the value of the attribute. The reader is referred to [47] for a more detailed explanation of the fast feature selection algorithm that is employed in this study.

3.2.2. LASSO

Least absolute shrinkage and selection operator (LASSO) is a method of regression analysis that can perform feature selection and regularization in machine learning models [48]. It is a very useful tool as it enhances the prediction accuracy and interpretability of the data mining model. In general, lasso regression has the objective of generating a subset of feature that minimizes the sum of squared errors with an upper bound on the sum of the absolute values of the model parameters. As a result of this upper bound, some of the parameters of the variables shrinks to zero. The variables with coefficients of zero are then excluded from the model, while variables with non-zero coefficients indicate a high association between the target and independent variables. For a detailed explanation of LASSO regression analysis method, the reader is referred to Tibshirani [48].

3.2.3. COX

Cox regression, also sometimes referred to as proportional hazards regression, is a regression method used to investigate the effect of various variables on the time a specific event occurs. In the context of this study, cox regression is useful for survival analysis. A detailed explanation about Cox regression can be found in Lin [49].

3.3. Sampling methods

The dataset used in this study generated an imbalanced dependent variable, for all of the three time points, as can be seen in Table 2. Imbalanced datasets can lead most classification models to classify most cases the same as the majority class in the training data. Various methods have been outlined in the machine-learning literature [50–52] to address this data imbalance problem. Contemporary methods approach the problem by resampling the dataset (Synthetic Minority Oversampling Technique (SMOTE), Random Under Sampling (RUS), and Adaptive Synthetic Sampling Approach (ADASYN) etc.). Resampling can either be used to over-sample the minority class or under-sample the majority class. There is no consensus in the machine learning literature with regards to the best approach for resampling data [50–54]. The best approach usually depends on the problem being solved as well as the dataset used for the problem. In this study, RUS is employed in order to overcome data imbalance problem. The majority classes are randomly sampled to contain a similar number of data points as the minority class. The rationale for us to employ the RUS, but not an oversampling technique is that RUS gave us very comparable results with the over-sampling techniques. Therefore, there was not a justifiable reason to create artificial points in the dataset. To address class imbalance while preventing data leakage and ensuring unbiased model evaluation, Random Under-Sampling (RUS) was fully integrated within the cross-validation framework. Specifically, for each fold of the 5-fold cross-validation process, RUS was applied exclusively to the training subset of that fold, while the corresponding test subset was left completely untouched to preserve the natural class distribution. The predictive model was then trained on the resampled training data and evaluated on the original test data for that fold. This fold-wise resampling strategy prevents information leakage from the test data into the training process and yields more reliable, reproducible, and generalizable performance estimates, consistent with established best practices in machine-learning methodology.

3.4. Cost-matrix models

Cost matrix is commonly used for machine learning classification models when there is an imbalance between the majority and minority class. In this approach, a penalty component could be included to avoid overfitting issues. In this study, survival and death cases were imbalanced. In the first timestamp (i.e. 1-month after transplantation surgery) most of people may survive (23,772 survivals versus 1165 deaths), while in the long term there are increases in death incidents for the patient population. To avoid biased predictions, *False Negatives (FP)* were penalized at a much higher rate when compared to *False Positives (FN)*. The penalization coefficient is assigned by taking the approximate value of dividing *survival* cases by *death* cases. Hence, penalization coefficients in this study are 20 (~23772/1165), 7.5 (19704/2693), and 2 (9788/4956) for 1-month, 1-year, and 5-year survival, respectively.

3.5. Prediction models

In this study, five classification algorithms such as *Tree Augmented Naïve Bayes (TAN)*, *Decision Trees (DT), Random Forests (RF)*, *Support Vector Machines (SVM) and Logistic Regression (LR)* were employed to predict the binary outcomes. These classification algorithms have been selected due to their superior performance in our preliminary analysis. What follows is the very brief discussion of these classification algorithms.

3.5.1. Tree augmented naïve (TAN) Bayesian belief network

A Bayesian Belief Network is a directed acyclic graph by which a probabilistic dependency model is represented. In such graph, the nodes represent the predictors while the conditional dependencies among the predictors are represented by the arcs [55,56]. It is commonly used for the situations where there might be complex nonlinear interactions among the predictors [57]. Tree Augmented Naïve Bayes (TAN) is a branch of Bayesian family where the target does not have any parents, while it is one of the parents of each predictor along with another variable. More specifically, a predictor variable might have at most two parents of which one of them is the target/outcome. In TAN model, an arc from X_1 to X2 means that, the contribution of X2 in predicting the target outcome is dependent on the value that X1 receives. The classification decision is made via Bayes rule where, for each predictor, the outcome probabilities are calculated. In the final phase, among these probabilities, the highest one is chosen for the structure [58]. For a more detailed description on TAN, we refer to reader to Dag et al. [3].

3.5.2. Logistic regression (LR)

Logistic regression is a standard regression method that are employed for either binary or multinomial classification problems. In such setting, the log odds of the dependent variable are modeled by linear combination of the independent predictors [59,60]. Logistics regression has been very commonly used for classification problems in the relevant medical literature [11,61,62] due to its "easy to interpret" structure.

3.5.3. Support vector machines (SVM)

Support vector machines (SVM), sometimes referred to as support vector regression, is a popular tool used for classification and regression in machine learning. It is one of the techniques used in solving big data classification problems in machine learning. A key drawback of SVM is that it is mathematically complex and also computationally expensive. Suthaharan [63] provides a detailed explanation of the SVM approach using process diagrams and data flow diagrams to illustrate the process, while Wee et al. [64] presents an application of the approach to predict linear B-cell epitopes.

3.5.4. Random forests

Random Forests method is a data mining algorithm that have been commonly used for both classification and regression problems [65] in many different fields as well as variable selection purposes in the recent literature. The prediction decision for the random forests is made based on either the *mode* (for classification) or the *mean* (regression) of the aggregated votes that are obtained by a collection of decision trees. Each tree in the forest is trained via a bootstrap sample of individuals. The construction procedure of each tree can be summarized through the following steps presented by Breiman [65];

Suppose there are I individuals and M explanatory attributes in the dataset,

1. A random sample whose size is equal to the training set is selected from the entire data.
2. At each node in the tree, m random attributes out of M *(*entire set of attributes in the data*)* is selected, where M/m *is* relatively much greater than *1*.
3. The best split is then chosen from the subset of m attributes selected in the previous step.
4. Second and third steps are iterated until there is fully grown tree is built (no pruning).

3.5.5. Decision trees (DT)

Decision trees (DT) are supervised learning technique commonly used in solving classification problems in machine learning. The overarching goal of a DT is to create a training model to predict the class or value of a target variable learning the simple decision rules stipulated for the model. Generally, DT develops a set of rules to split the data into the smaller fractions based on their homogeneity. The rules are a set of questions that drive the parent point into the children until it reaches to the leaves that represent the classes' labels. Timofeev [66] provides a detailed explanation of the application of decision trees.

4. Results and discussion

4.1. Variable selection results via IG-FFS, RF and literature review

Recall that three feature selection models (i.e., *IG-FFS, LASSO* and *COX* models) were employed in selecting the variables, in addition to the variables that were determined by an extensive literature review. It should also be noted that there are three different time points (i.e., 1-month, 1-, and 5-years) that have been investigated to see whether the newly added variables have any meaningful contribution in predicting the *gstatus* in different time horizons. The number of the features that were found to be "good" via the feature selection algorithms as well as the variables that were found to be (potentially) important in the existing literature are presented in Table 3.

Based on Table 3, for the 1-month time stamp, it can be seen that there are 1, 2, 2 and 4 variables from the newly added variables in 2004 (9 in total) selected through *IG-FFS, LASSO, COX,* and literature review analysis, respectively. Because of the overlap among these variable sets there are only 5 (newly added) variables (in total) to be deployed into the classification algorithms. A similar pattern can be seen for 1- and 5-year analysis as well. It should be noted that there are 22 (17 + 5), 39

Table 3
The number of the features selected through variable selection methods and literature review.

	IG-FFS		LASSO		COX		Lit. Rev		Total	
	Old	New	Old	New	Old	New	Old	New	Old	New
1-month	4	1	11	2	5	2	11	4	17	5
1-year	5	1	15	1	5	2	23	5	33	6
5-year	5	1	31	3	5	2	27	6	53	7

(33+6) and 60 (53+7) variables to be deployed in the final 1-month, 1-year and 5-year prediction models, respectively.

In Table 4, the new variables, which were selected through the comprehensive selection procedure, are presented for each time point in detail. The rationale behind presenting only the new variables (but not the old ones) in Table 4 is that the main goal of this study is to specifically focus on the newly added variables. In this table × and $\surd$ denote whether the variable is selected through the associated variable selection model or not (with × representing "No" and $\surd$ representing "Yes"). It should be noted that as long as a feature is selected through any of the selection methods, it is to be included in the prediction models (in the next phase). Therefore, column "Inclusion" denotes whether the variables will be included in the prediction models or not. As it also can be confirmed by Table 4 that there are 5, 6, and 7 variables that will be included (selected through at least one of the feature selection methods) in the 1-month, 1-year and 5-year time predictions, respectively. In the next subsection, the prediction (classification) results obtained through employing the selected (old + new) variables for three different time points are presented in detail.

4.2. Classification results

Based on the discussion in Section 3.5, TAN, LR, SVM, DT, RF and LR were employed to predict the *gstatus* (outcome) for three different time points (i.e., 1-month, 1-year, and 5-years) by both excluding and including the newly added variables in the prediction models using the 5-fold cross validation concept. The rationale behind employing cross validation concept is to decrease the bias in the outcomes as well as to increase the robustness of the classification models [67]. Therefore, we have 5 (classification models) X 2 (for old and combined set of variables) X 3 (2 different sampling methods (i.e., *RUS* and *NO*) + 1 cost matrix model) X 3 (different time points) X 5 (five-fold cross validation) = 450 different models to run in total.

The summary of the classification results with highest AUC that were obtained through using these models for each time frame are presented in Table 6. It should also be noted that the extended results including all the combinations of models, resampling, and time stamps are presented in APPENDIX I. *AUC (area under the Curve), accuracy*, *sensitivity*, and *specificity* metrics were presented for all models as the evaluation criteria. For a more detailed description of these criteria, the reader is referred to Fawcett [68]. *AUC* will be used as the main evaluation criterion among these metrics since it is considered to be an objective criterion specifically in evaluating the classification performances for imbalanced datasets [50,69] as in our case. Having said that, models with extreme sensitivity values are also not considered even if their *AUC* scores are reasonable. The rationale behind that is that these models have automatically labeled every patient as "*failed*" because of the extreme imbalance in the data (e.g., in *1-month* and *1-year* situation). As can be seen from Table 6, for each time frame, same classification models with and without the newly added variables were compared. To exemplify, the best performing LR model with the "old" variables were compared against the best performing LR model with the "combined" variables. Same procedures were applied to all the five classification models that were employed in the proposed study. To explain it in more detail, let us look at the 1-month values for LR model. It can be noted that LR model, when "old" variables are employed only, performed best when the classification cost matrix approach was employed in the training stage of the model, and so on and so forth.

One of the most important observations that can be made is from Table 6 is that SVM models (for all the three-time frames) outperformed the other four classification algorithms. Another observation that can be made is that the no-sampling usually produces very imbalanced results between the sensitivity and specificity measure except the DT model. It is also evident that, except the DT models, either RUS sampling or Cost matrix-based (CM) models produce the best AUC results for all the time horizons. Needless to say, RUS and CM reduce the gap between the sensitivity and specificity results.

Another observation that can be drawn from Table 6 is that there is no substantial evidence that addition of 2004 variables could increase predictive performance for the 1-month and 1-Year time stamps, while this addition could improve prediction performance for the fifth year. Except some of the 5-year time horizon numbers, the newly introduced variables did not contribute to improving the prediction performance. In fact, 8 (out of 10) comparisons in 1-month and 1-year time frames, the classification models slightly underperformed when the new variables are included. However, the differences between the "old" and "combined" are usually very small (< 0.01) that some of them can be ignored. For 5-year though, the results are "*tie*", meaning sometimes the new variables contribute and sometimes not. Having said that, the differences are again very small (< 0.01).

To ensure a conservative and robust statistical evaluation, we adopted a stringent significance threshold of $\alpha = 0.01$ for all hypothesis testing. This choice was motivated by the large number of model comparisons performed across multiple algorithms and prediction horizons (1-month, 1-year, and 5-year), which increases the risk of false-positive findings when conventional thresholds (e.g., $p < 0.05$) are used. By applying a stricter criterion, we aimed to minimize Type I error and ensure that only highly consistent and reproducible performance differences are interpreted as statistically meaningful. This conservative approach aligns with best practices in predictive modeling studies involving multiple comparisons and strengthens the reliability of our statistical conclusions.

Using this conservative framework, paired t-test results based on fold-wise AUC values indicate that the inclusion of the newly added variables does not lead to a statistically significant improvement in predictive performance for the vast majority of models. Although a small number of isolated comparisons reach significance at the conventional 0.05 level, these effects do not persist under the stricter 0.01 threshold and are not consistent across time horizons. Furthermore, the limited instances of statistical significance—primarily observed in Decision Tree models—do not demonstrate a systematic or reproducible performance gain attributable to the additional variables. Collectively, these findings provide strong statistical evidence that the newly introduced predictors do not yield robust or generalizable improvements in model discrimination, supporting the central conclusion of this study.

Henceforth, models with the highest AUC in each time stamp is selected for further discussion and the analysis in the next subsection (sensitivity analysis). To exemplify, the *LR_CM(old), RF_RUS(old), SVM_CM(old), TAN_RUS(old), DT_RUS(combined)* are used in the next section (*Sensitivity Analysis)* for the 1-month time horizon and etc.

4.3. Sensitivity analysis

Sensitivity analysis is an alternative common measure in the data mining community [70] that enables one discover the relative contribution of each variable towards predicting the outcome variable (*gstatus*). In such setting, a predictive variable is found to be important if its addition results in improving the model's prediction performance, as measured by the *AUC* metric. According to the sensitivity measure defined by Saltelli et al. [70], the sensitivity measure is defined as

$$S_i = \frac{V_i}{V_y} = \frac{V(E(y/x_i))}{V(y)} \tag{4}$$

Table 4
(Newly added) Variables that are selected through different time points.

	1-month					1-year				5-year			
	COX	IG-FFS	LASSO	Lit. Rev	Inclusion	IG-FFS	LASSO	Lit. Rev	Inclusion	IG-FFS	LASSO	Lit. Rev	Inclusion
ALCOHOL_HEAVY_DON													
ARGININE_DON				✔	✔			✔	✔			✔	✔
BMI_TCR								✔	✔		✔	✔	✔
CDC_RISK_HIV_DON				✔	✔			✔	✔		✔	✔	✔
CIG_USE	✔				✔				✔				✔
HEMATOCRIT_DON													
INSULIN_DON													
INOTROP_VASO_CO_TCR													
INOTROP_VASO_DIA_TCR													
INOTROP_VASO_MN_TCR													
INOTROP_VASO_PCW_TCR													
PCO2_DON													
PH_DON													
TCR_DUR_ABSTAIN													
TRANSFUS_TERM_DON												✔	✔
VAD_DEVICE_TY_TCR	✔		✔	✔	✔			✔	✔			✔	✔
VAD_DEVICE_TY_TRR		✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
WORK_INCOME_TCR													

Table 5
The list and brief description of the classification models used for each time-period.

Model	Description
LR_RUS (old)	LR sampled with RUS by including only the old variables
LR_CM (old)	LR with Cost Matrix model including only the old variables
LR_NO (old)	LR without re-sampling by including only the old variables
LR_RUS (combined)	LR sampled with RUS by including both the new and the old variables
LR_CM (combined)	LR with Cost Matrix model including both the new and the old variables
LR_NO (combined)	LR without re-sampling by including both the new and the old variables
DT_RUS (old)	DT sampled with RUS by including only the old variables
DT_CM (old)	DT with Cost Matrix model including only the old variables
DT_NO (old)	DT without re-sampling by including only the old variables
DT_RUS (combined)	DT sampled with RUS by including both the new and the old variables
DT_CM (combined)	DT with Cost Matrix model including both the new and the old variables
DT_NO (combined)	DT without re-sampling by including both the new and the old variables
RF_RUS (old)	RF sampled with RUS by including only the old variables
RF_CM (old)	RF with Cost Matrix model over Survive including only the old variables
RF_NO (old)	RF without re-sampling by including only the old variables
RF_RUS (combined)	RF sampled with RUS by including both the new and the old variables
RF_CM (combined)	RF with Cost Matrix model over Survive including both the new and the old variables
RF_NO (combined)	RF without re-sampling by including both the new and the old variables
SVM_RUS (old)	SVM sampled with RUS by including only the old variables
SVM_CM (old)	SVM with Cost Matrix model including only the old variables
SVM_NO (old)	SVM without re-sampling by including only the old variables
SVM_RUS (combined)	SVM sampled with RUS by including both the new and the old variables
SVM_CM (combined)	SVM with Cost Matrix model including both the new and the old variables
SVM_NO (combined)	SVM without re-sampling by including both the new and the old variables
TAN_RUS (old)	TAN sampled with RUS by including only the old variables
TAN_CM (old)	TAN with Cost Matrix model including only the old variables
TAN_NO (old)	TAN without re-sampling by including only the old variables
TAN_RUS (combined)	TAN sampled with RUS by including both the new and the old variables
TAN_CM (combined)	TAN with Cost Matrix model including both the new and the old variables
TAN_NO (combined)	TAN without re-sampling by including both the new and the old variables

where y is the dichotomous output variable (*gstatus*), and the unconditional output variance is denoted by $V(y)$. The expectation operator is denoted by E, which calls for an integral over all predictor variables except x_i. A further integral operator is implied over x_i by the operator V_i. The importance of a specific variable is then computed as the normalized sensitivity.

In this subsection, sensitivity analysis is performed on the models with the highest AUC in each time stamps (i.e., 1-month, 1-year, 5-year) based on the results discussed in Section 4.2. It should be heavily emphasized that the Figs. 2–6 should be read along with Tables 3, 4 and 5 for the sake of comprehending the holistic picture.

Fig. 2 represents the 15 variables that are found to be the most contributory ones in predicting the 1- month survival after cardiac transplant. As can be seen from the figure, there are no "new" variables among the most important 15 variables that predict the 1-month survival. In other words, none of the five variables that were checked under the "inclusion" column in Table 4 (i.e. *ARGININE_DON, CDC_RISK_HIV_DON, CIG_USE, VAD_DEVICE_TY_TCR*, *VAD_DEVICE_TY_TRR*) were among the most important 15 variables. This makes sense when 1-month results are considered in Table 6.

Fig. 3 represents the 15 variables that are found to be the most contributory ones in predicting the 1- year survival after cardiac transplant. It should be noted that, similar to 1-month results, there are no "new" variables among the most important 15 variables that predict the 1-year survival. In other words, none of the six variables that were checked under the "inclusion" column in Table 4 happened to be among the most important 15 variables. This also makes sense when the associated results are considered in Table 6.

As can be seen in Fig. 3, 1-year post-transplant survival is jointly shaped by recipient clinical condition, procedural complexity, and donor characteristics. Indicators of end-organ dysfunction, including the level of creatinine (CREAT_TRR) and bilirubin (TBILI). Likewise,

Table 6
Classification results obtained through including and excluding the newly added variables.

Time horizon	Model	# Variables	AUC	Sensitivity	Specificity	Accuracy	p-values
1-month	LR_CM (old)	17	0.6721	0.6944	0.5479	0.6868	0.0494
1-month	LR_CM (combined)	17+5	0.6662	0.7087	0.5249	0.699	
1-month	RF_RUS (old)	17	0.6564	0.6086	0.6245	0.6094	0.0272
1-month	RF_RUS (combined)	17+5	0.6455	0.6147	0.5709	0.6124	
1-month	SVM_CM (old)	17	0.6721	0.7144	0.5364	0.705	0.0507
1-month	SVM_CM (combined)	17+5	0.6628	0.7199	0.5096	0.7089	
1-month	TAN_RUS (old)	17	0.6183	0.5577	0.6054	0.5602	0.3895
1-month	TAN_RUS (combined)	17+5	0.6156	0.5931	0.5785	0.5923	
1-month	DT_NO (old)	17	0.5848	0.7392	0.3985	0.7213	**0.0000**
1-month	DT_RUS (combined)	17+5	0.609	0.6198	0.5977	0.6186	
1-year	LR_CM (old)	33	0.6337	0.6508	0.5472	0.6379	0.0272
1-year	LR_RUS (combined)	33+6	0.6278	0.6251	0.5508	0.6158	
1-year	RF_RUS (old)	33	0.6243	0.585	0.6007	0.587	0.1813
1-year	RF_RUS (combined)	33+6	0.6174	0.6036	0.5544	0.5975	
1-year	SVM_RUS (old)	33	0.6342	0.6263	0.549	0.6167	0.0158
1-year	SVM_RUS (combined)	33+6	0.6285	0.6082	0.5722	0.6037	
1-year	TAN_CM (old)	33	0.5899	0.6988	0.4367	0.666	0.1190
1-year	TAN_CM (combined)	33+6	0.584	0.7141	0.4296	0.6785	
1-year	DT_RUS (old)	33	0.5821	0.4063	0.6988	0.443	**0.0006**
1-year	DT_NO (combined)	33+6	0.6049	0.5929	0.5722	0.5903	
5-year	LR_CM (old)	53	0.6266	0.6262	0.5466	0.5987	0.0255
5-year	LR_CM (combined)	53+7	0.622	0.6185	0.5486	0.5943	
5-year	RF_RUS (old)	53	0.62	0.6065	0.5751	0.5957	0.0873
5-year	RF_NO (combined)	53+7	0.6303	0.9269	0.1776	0.6679	
5-year	SVM_CM (old)	53	0.6365	0.621	0.579	0.6065	0.3508
5-year	SVM_CM (combined)	53+7	0.6385	0.6029	0.5957	0.6004	
5-year	TAN_RUS (old)	53	0.6071	0.5993	0.5702	0.5892	0.3702
5-year	TAN_RUS (combined)	53+7	0.6051	0.5931	0.5613	0.5821	
5-year	DT_NO (old)	53	0.5702	0.6941	0.422	0.6001	1.0000
5-year	DT_NO (combined)	53+7	0.5702	0.6941	0.422	0.6001	

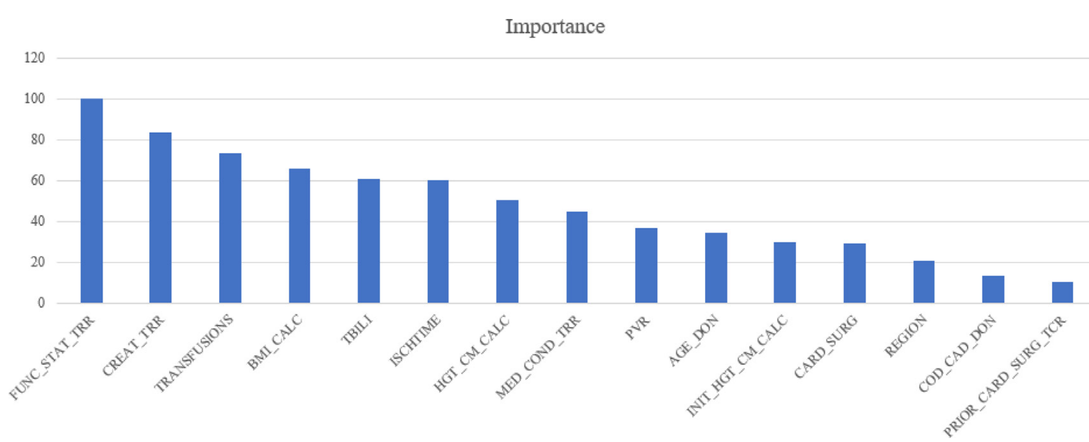

Fig. 2. Top 15 most contributory predictors for 1-month survival prediction.

amount of serum albumin (TOT_SERUM_ALBUM) serves as a marker of post-transplant recovery. A history of intravenous drug use or active infection (INFECT_IV_DRUG_TRR) were also found to be associated with survival. Functional status at the time of transplantation (FUNC_STAT_TRR) emerges as one of the strongest predictors. Procedural factors also contribute substantially, such as prolonged ischemic time (ISCHTIME) and whether the patient has had prior cardiac surgeries (CARD_SURG). In addition, Donor-related characteristics further modulate outcomes, such as donor age (AGE_DON) and donor–recipient gender mismatch (GENDER_DON). Finally, the specific type of prior cardiac surgery (PRIOR_CARD_SURG_TYPE_TCR) refines overall risk stratification. Collectively, these factors interact to define recipient resilience, donor organ quality, and surgical complexity, ultimately determining 1-year post-transplant survival.

Building on this conceptual framework, Fig. 4 further enables the identification of clinically meaningful conditional independence relationships among predictors. This proposed DAG (Directed Acyclic Graph) was inferred from the underlying Tree-Augmented Naïve Bayes (TAN) framework, which captures conditional dependencies among predictors beyond the naïve independence assumption. For example, once intravenous drug use or infection status is known (INFECT_IV_DRUG_TRR), renal function (CREAT_TRR) becomes independent of donor gender (GENDER_DON), indicating that their apparent association is largely mediated through infection-related pathways rather than a direct biological effect. Similarly, conditioning on infection status renders creatinine (CREAT_TRR) and serum albumin (TOT_SERUM_ALBUM) approximately independent, as severe infections tend to concurrently impair renal function and nutritional reserve. In

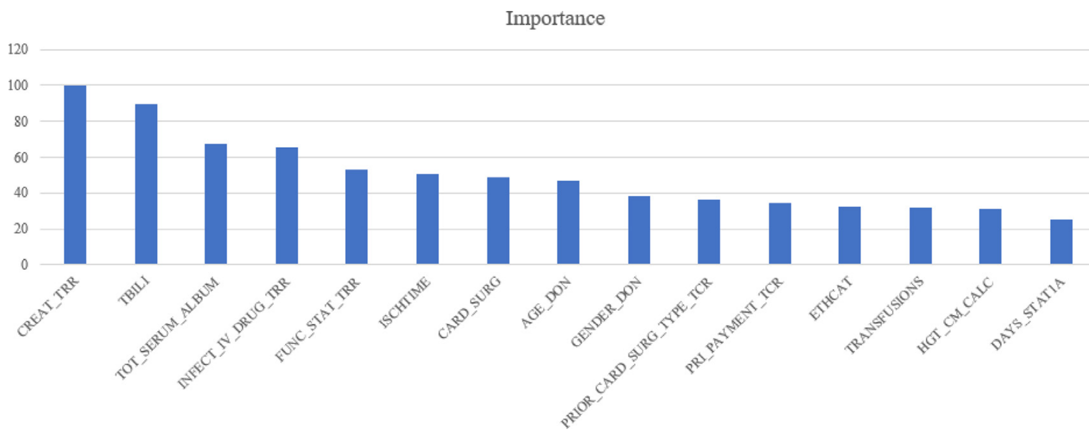

Fig. 3. Top 15 most contributory predictors for 1-year survival prediction.

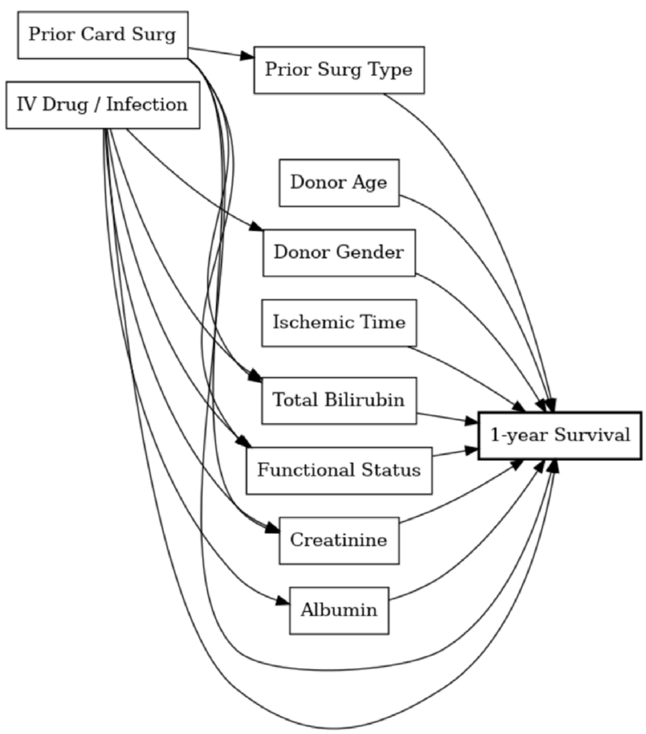

Fig. 4. Conditional relations among 10 most contributory predictors for 1-year survival prediction.

addition, functional status (FUNC_STAT_TRR) becomes independent of laboratory markers such as creatinine, bilirubin (TBILI), and albumin after accounting for infection history (INFECT_IV_DRUG_TRR) and prior cardiac surgery (CARD_SURG), suggesting that these upstream clinical factors largely determine both physiological reserve and end-organ function. Moreover, donor age (AGE_DON) remains independent of recipient laboratory profiles, reflecting that donor characteristics are primarily driven by organ availability and allocation rather than recipient health status. Taken together, these conditional independence structures demonstrate how the DAG not only captures direct effects on survival but also disentangles spurious associations by conditioning on common causes, reinforcing the clinical plausibility of the model and highlighting its utility in clarifying complex dependency patterns among transplant-related risk factors.

Similarly, Fig. 5 depicts the 15 variables that are found to be the most contributory ones in predicting the 5-year survival. *ARGININE_DON and BMI_TCR* were the only two "new" variables that were selected to be among the top 15 variables. According to the little information available in the existing literature [71], the use of low dose arginine vasopressin (*ARGININE_DON*), which is used for retaining water in the body and constricting blood vessels), can be used to support brain-dead thoracic organ donors. Donor arginine (*ARGININE_DON)* status may play an important role in long-term post-heart transplant outcomes through its effects on endothelial function and vascular homeostasis. Arginine is a key substrate for nitric oxide (NO) synthesis, which is essential for maintaining vascular tone, inhibiting platelet aggregation, and reducing inflammatory responses within the graft vasculature. Adequate nitric oxide bioavailability has been associated with improved endothelial integrity and protection against chronic allograft vasculopathy, a major determinant of long-term graft survival. While early post-transplant outcomes are primarily driven by perioperative factors such as ischemia-reperfusion injury, acute rejection, and surgical complications, longer-term survival is more strongly influenced by progressive vascular remodeling and chronic inflammatory processes. In this context, donor arginine status may exert a delayed but clinically meaningful effect by modulating endothelial health over time, which could explain its association with 5-year survival but not with short-term (e.g., 1-month) mortality. These findings suggest that metabolic and endothelial-related donor characteristics may have greater relevance for long-term graft durability rather than immediate post-operative risk.

Recipient body mass index (BMI_TCR) may influence long-term post-heart transplant outcomes through its impact on metabolic status, inflammatory burden, and cardiovascular reserve. Elevated BMI is associated with chronic low-grade inflammation, insulin resistance, and endothelial dysfunction, all of which can adversely affect graft function over time. In addition, higher BMI has been linked to impaired immune regulation and altered pharmacokinetics of immunosuppressive therapies, potentially increasing the risk of chronic rejection and graft vasculopathy.

Similar to donor arginine, the clinical impact of BMI_TCR may not be immediately evident in the early post-transplant period, where outcomes are largely driven by surgical factors, acute rejection episodes, and perioperative complications. However, over longer follow-up horizons, persistent metabolic stress and systemic inflammation associated with higher BMI may contribute to progressive vascular remodeling, graft dysfunction, and reduced long-term survival. This may explain why BMI_TCR emerges as an important predictor for longer-term outcomes (e.g., 5-year survival) but not for short-term mortality.

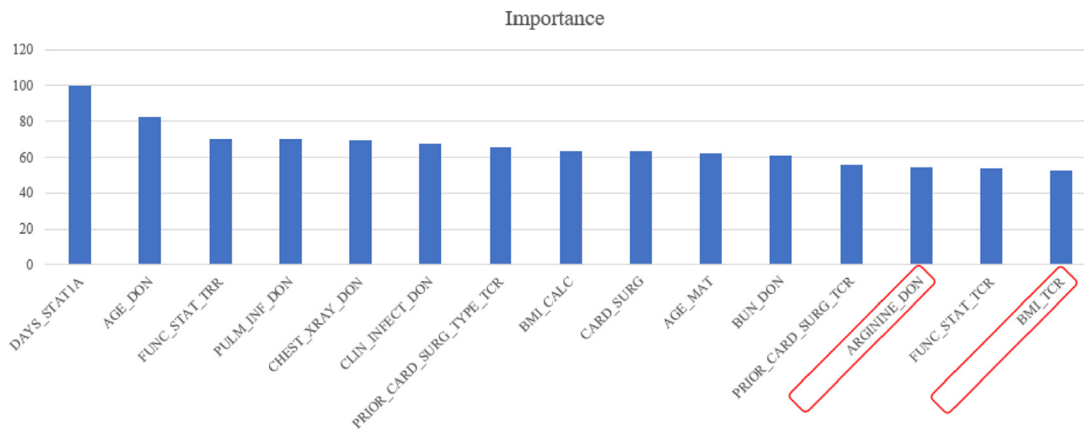

Fig. 5. Top 15 most contributory predictors for 5-year survival prediction.

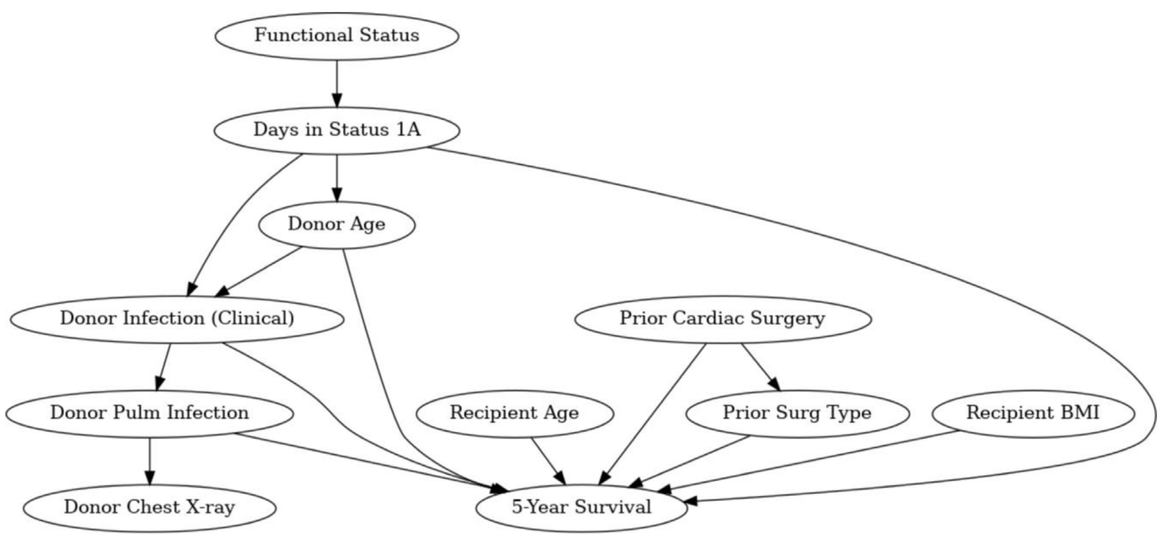

Fig. 6. Conditional relations among 10 most contributory predictors for 5-year survival prediction.

These findings highlight the importance of recipient metabolic health in determining long-term graft durability and support the role of BMI as a clinically meaningful predictor in transplant risk stratification.

As can be inferred from Fig. 6, we present a structured framework for analyzing 5-year post-transplant survival using ten key variables encompassing donor characteristics, recipient clinical status, and pre-transplant urgency. This binary outcome is jointly influenced by donor factors such as age and infection status, as well as recipient-related factors including age, body mass index, functional status, and prior cardiac surgeries. The proposed DAG captures the underlying dependency structure, illustrating how advanced donor and recipient age, prolonged urgent wait times, poor functional status, and donor infections collectively shift outcomes toward lower long-term survival. Moreover, the model highlights important conditional independence relationships, demonstrating how certain predictors become redundant once upstream clinical information is accounted for.

In the donor infection subnetwork, abnormal donor chest X-ray findings (CHEST_XRAY_DON) become independent of survival once donor pulmonary infection status is known, indicating that chest X-ray abnormalities (CHEST_XRAY_DON) act primarily as indicators of underlying infection rather than direct risk factors. Similarly, the specific type of prior cardiac surgery (PRIOR_CARD_SURG_TYPE_TCR) is conditionally independent of survival after accounting for whether the patient had any prior surgery (CARD_SURG), suggesting that the overall surgical history captures most of the associated risk. The DAG also illustrates a collider effect at the survival node, whereby risk factors such as donor age (AGE_DON) and recipient functional status (FUNC_STAT_TRR) are marginally independent but become conditionally dependent when survival status is known, reflecting an "explaining-away" phenomenon common in Bayesian networks. Finally, a mediator structure is observed in the pathway from functional status to survival through urgency status (DAYS_STAT1 A), indicating that functional status influences long-term outcomes largely through its impact on transplant urgency and waiting time. Together, these patterns demonstrate how the DAG disentangles direct effects, mediating pathways, and spurious associations, reinforcing its value for interpreting complex dependency structures in long-term transplant outcomes. Finally, recipient age and donor–recipient age matching (AGE_MAT) play a critical role.

Overall, five-year survival after heart transplantation is a multifactorial process. The Bayesian network framework facilitates disentangling direct and indirect effects, identifies mediating pathways (e.g., functional status acting through urgency), and clarifies which variables provide incremental predictive value. This approach parallels our 1-year analysis while extending it to long-term outcomes, where chronic patient characteristics and cumulative surgical burden play a more prominent role. Ultimately, this framework offers a principled strategy for risk stratification and supports more informed clinical decision-making to improve long-term transplant outcomes.

5. Conclusions, future research recommendations and limitations

Data collection is an essential feature of the healthcare sector. This is done both through automated data collection mechanisms and manual data collection methods such as taking blood samples, running lab tests and medical imaging. Irrespective of the method used, data collection is a time consuming and expensive process not only in healthcare but also in other fields. The primary goal of this research was to conduct a rigorous evaluation of the variables added to the UNOS heart transplant database after June 30, 2004. By focusing specifically on survival life (SL) prediction—spanning short-, mid-, and long-term horizons—this study addresses the critical trade-off between the costs of data collection (manual labor, capital investment, and computational complexity) and the clinical benefits of enhanced predictive accuracy. In addition, the study examined the value of collecting additional data in enhancing prediction accuracy.

To evaluate these variables, we proposed a novel five-phase data analytical framework:

(a) Data Preparation: Cleaning and normalizing the expanded UNOS dataset.
(b) Feature Selection: Utilizing algorithms to identify which of the post-2004 variables held the highest predictive value.
(c) Prediction Modeling: Employing five distinct data mining-based approaches to test performance improvements.
(d) Sensitivity Analysis: Measuring the individual contribution of selected features to the model's output.
(e) Clinical Evaluation: Assessing the impact of these results on medical decision-making.

The study sought to answer three pivotal questions regarding the 2004 data expansion:

(a) Feature Importance: Which of the newly added variables were identified to be important by the feature selection method, thereby justifying their continued collection in the context of survival modeling.
(b) Predictive Power: Does the power of the prediction model (prediction performance) improve by including the new variables that were selected using the variable selection mechanism?
(c) Variable Contribution: What is the contribution of each of these (new) variables in predicting the gstatus for different time horizons?

From a clinical perspective, this framework allows for superior risk stratification. By identifying "high-risk" patients through these enhanced models, medical practitioners can tailor post-transplant care—adjusting follow-up frequencies and intensifying monitoring for complications such as Coronary Allograft Vasculopathy (CAV), renal dysfunction, and thromboembolic events.

Although the newly introduced variables may hold potential value in upstream decision-making processes such as donor–recipient matching, the present study intentionally focuses on post-transplant survival prediction. These two applications require fundamentally different modeling frameworks, outcome definitions, and validation strategies. Accordingly, we emphasize that our findings do not negate the possible utility of these variables in allocation or matching contexts, but rather demonstrate that their incremental contribution to downstream survival prediction is limited within the modeling framework examined here. Future studies are warranted to explicitly investigate their role in donor–recipient matching and transplant allocation strategies.

Contrary to the perception that null results lack novelty, our findings provide important negative evidence that challenges a commonly held assumption in predictive modeling—that adding more variables necessarily improves performance. By rigorously demonstrating that this expected improvement does not materialize, even under generous feature inclusion and robust validation, our study contributes actionable methodological insight and discourages unnecessary model complexity.

While this study focused strictly on heart transplant survival, the proposed methodology is highly scalable. This data analytical approach can be extended to other organ transplant datasets provided by UNOS (e.g., kidney, liver, or lung), facilitating a more data-efficient and clinically effective transplant ecosystem.

In addition, several limitations should be acknowledged when interpreting the results of this study. First, although the UNOS registry provides a large and comprehensive dataset, it may contain potential biases or incompleteness inherent to registry-based data, which could affect model training and outcome estimation. Second, the absence of external validation limits the ability to fully assess the generalizability of the proposed models to independent populations and real-world clinical settings. Third, feature-selection procedures may be sensitive to data imbalance and preprocessing choices, which could influence the stability and selection frequency of predictors across different cross-validation folds. Finally, some of the observed differences in model performance may fall within expected sampling variability rather than reflecting true methodological superiority. Recognizing these considerations strengthens the interpretation of our findings and aligns this work with standard expectations for predictive modeling studies.

Last but not least, while this study leverages a large and well-established dataset, it is important to note that the transplant cases included extend up to 2012. Since then, notable advancements have occurred in heart transplantation, particularly in surgical techniques and post-operative management, including improvements in mechanical circulatory support (MCS) devices. As a result, model performance observed in this study may not fully capture outcomes under the most current clinical practices. Nevertheless, the dataset provides valuable historical insight and a strong foundation for predictive modeling in this domain.

From a managerial standpoint, this study provides a blueprint for cost–benefit analysis in healthcare data. It assists organizations in deciding where to focus expensive data collection efforts, ensuring that investments are made in variables that truly move the needle for patient outcomes and quality of life (QOL).

Declaration of competing interest

The authors have no relevant financial or non-financial interests to disclose and no competing interests to declare that are relevant to the content of this article.

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Appendix

See Table A.1.

Data availability

Data will be made available on request.

Table A.1
Prediction results of all combinations of models, time stamps and resampling methods.

		# of variables	AUC	Sensitivity	Specificity	Accuracy
1- Month	LR_RUS (old)	17	0.6606	0.6535	0.5747	0.6494
	LR_CM (old)	17	0.6721	0.6944	0.5479	0.6868
	LR_NO (old)	17	0.6680	0.9994	0.0038	0.9472
	LR_RUS (combined)	(17+5)	0.6561	0.6518	0.5594	0.6470
	LR_ CM (combined)	(17+5)	0.6662	0.7087	0.5249	0.6990
	LR_NO (combined)	(17+5)	0.6644	0.9992	0.0038	0.9470
	RF_RUS (old)	17	0.6564	0.6086	0.6245	0.6094
	RF_ CM (old)	17	0.6211	0.9972	0.0115	0.9455
	RF_NO (old)	17	0.6338	1.0000	0.0000	0.9476
	RF_RUS (combined)	(17+5)	0.6455	0.6147	0.5709	0.6124
	RF_ CM (combined)	(17+5)	0.6443	0.9989	0.0000	0.9466
	RF_NO (combined)	(17+5)	0.6423	0.9996	0.0000	0.9472
	SVM_RUS (old)	17	0.6505	0.6794	0.5441	0.6723
	SVM_ CM (old)	17	0.6721	0.7144	0.5364	0.7050
	SVM_NO (old)	17	0.6613	0.6969	0.5402	0.6887
	SVM_RUS (combined)	(17+5)	0.6512	0.6582	0.5287	0.6514
	SVM_ CM (combined)	(17+5)	0.6628	0.7199	0.5096	0.7089
	SVM_NO (combined)	(17+5)	0.5830	1.0000	0.0000	0.9476
	TAN_RUS (old)	17	0.6183	0.5577	0.6054	0.5602
	TAN_ CM (old)	17	0.5584	0.7322	0.3716	0.7133
	TAN_NO (old)	17	0.6226	0.9981	0.0038	0.9460
	TAN_RUS (combined)	(17+5)	0.6156	0.5931	0.5785	0.5923
	TAN_ CM (combined)	(17+5)	0.5459	0.7434	0.3563	0.7231
	TAN_NO (combined)	(17+5)	0.6256	0.9955	0.0192	0.9443
	DT_RUS (old)	17	0.5330	0.6514	0.4023	0.6383
	DT_ CM (old)	17	0.5555	0.5778	0.5172	0.5746
	DT_NO (old)	17	0.5848	0.7392	0.3985	0.7213
	DT_RUS (combined)	(17+5)	0.6090	0.6198	0.5977	0.6186
	DT_ CM (combined)	(17+5)	0.5657	0.5929	0.5019	0.5881
	DT_NO (combined)	(17+5)	0.5939	0.6575	0.5096	0.6498
1-Year	LR_RUS (old)	33	0.6336	0.6279	0.5704	0.6207
	LR_CM (old)	33	0.6337	0.6508	0.5472	0.6379
	LR_NO (old)	33	0.6449	0.9977	0.0143	0.8745
	LR_RUS (combined)	(33+6)	0.6278	0.6251	0.5508	0.6158
	LR_ CM (combined)	(33+6)	0.6269	0.6562	0.5169	0.6388
	LR_NO (combined)	(33+6)	0.6374	0.9977	0.0196	0.8752
	RF_RUS (old)	33	0.6243	0.5850	0.6007	0.5870
	RF_ CM (old)	33	0.6161	0.9954	0.0160	0.8727
	RF_NO (old)	33	0.6318	0.9987	0.0071	0.8745
	RF_RUS (combined)	(33+6)	0.6174	0.6036	0.5544	0.5975
	RF_ CM (combined)	(33+6)	0.6024	0.9931	0.0196	0.8712
	RF_NO (combined)	(33+6)	0.6335	0.9982	0.0089	0.8743
	SVM_RUS (old)	33	0.6342	0.6263	0.5490	0.6167
	SVM_ CM (old)	33	0.6072	0.8389	0.2763	0.7685
	SVM_NO (old)	33	0.5988	1.0000	0.0018	0.8750
	SVM_RUS (combined)	(33+6)	0.6285	0.6082	0.5722	0.6037
	SVM_ CM (combined)	(33+6)	0.6174	0.8441	0.2799	0.7734
	SVM_NO (combined)	(33+6)	0.6072	0.9997	0.0000	0.8745
	TAN_RUS (old)	33	0.5740	0.5421	0.5490	0.5430
	TAN_ CM (old)	33	0.5899	0.6988	0.4367	0.6660
	TAN_NO (old)	33	0.6059	0.9855	0.0321	0.8660
	TAN_RUS (combined)	(33+6)	0.5750	0.5539	0.5472	0.5530
	TAN_ CM (combined)	(33+6)	0.5840	0.7141	0.4296	0.6785
	TAN_NO (combined)	(33+6)	0.6020	0.9811	0.0642	0.8663

(continued on next page)

Table A.1 (*continued*).

	DT_RUS (old)	33	0.5821	0.4063	0.6988	0.4430
	DT_ CM (old)	33	0.5557	0.5592	0.5490	0.5579
	DT_NO (old)	33	0.5672	0.7846	0.3387	0.7287
	DT_RUS (combined)	(33+6)	0.5608	0.5222	0.5508	0.5258
	DT_ CM (combined)	(33+6)	0.5745	0.6623	0.4492	0.6356
	DT_NO (combined)	(33+6)	0.6049	0.5929	0.5722	0.5903
5-Year	LR_RUS (old)	53	0.6266	0.6185	0.5672	0.6007
	LR_ CM (old)	53	0.6266	0.6262	0.5466	0.5987
	LR_NO (old)	53	0.6284	0.9202	0.2051	0.6730
	LR_RUS (combined)	(53+7)	0.6216	0.6086	0.5711	0.5957
	LR_ CM (combined)	(53+7)	0.6220	0.6185	0.5486	0.5943
	LR_NO (combined)	(53+7)	0.6236	0.9108	0.2228	0.6730
	RF_RUS (old)	53	0.6200	0.6065	0.5751	0.5957
	RF_ CM (old)	53	0.6198	0.8808	0.2591	0.6659
	RF_NO (old)	53	0.6193	0.9264	0.1747	0.6666
	RF_RUS (combined)	(53+7)	0.6254	0.6153	0.5702	0.5997
	RF_ CM (combined)	(53+7)	0.6153	0.8823	0.2444	0.6618
	RF_NO (combined)	(53+7)	0.6303	0.9269	0.1776	0.6679
	SVM_RUS (old)	53	0.6297	0.6345	0.5770	0.6147
	SVM_ CM (old)	53	0.6365	0.6210	0.5790	0.6065
	SVM_NO (old)	53	0.6243	0.9663	0.1187	0.6733
	SVM_RUS (combined)	(53+7)	0.6299	0.6257	0.5653	0.6048
	SVM_ CM (combined)	(53+7)	0.6385	0.6029	0.5957	0.6004
	SVM_NO (combined)	(53+7)	0.6231	0.9601	0.1246	0.6713
	TAN_RUS (old)	53	0.6071	0.5993	0.5702	0.5892
	TAN_ CM (old)	53	0.5937	0.6418	0.5015	0.5933
	TAN_NO (old)	53	0.6043	0.8129	0.3229	0.6435
	TAN_RUS (combined)	(53+7)	0.6051	0.5931	0.5613	0.5821
	TAN_ CM (combined)	(53+7)	0.5920	0.6501	0.5093	0.6014
	TAN_NO (combined)	(53+7)	0.6044	0.8123	0.3425	0.6499
	DT_RUS (old)	53	0.5679	0.6661	0.4357	0.5865
	DT_ CM (old)	53	0.5607	0.5329	0.5505	0.5390
	DT_NO (old)	53	0.5702	0.6941	0.4220	0.6001
	DT_RUS (combined)	(53+7)	0.5597	0.4723	0.6251	0.5251
	DT_ CM (combined)	(53+7)	0.5610	0.7543	0.3533	0.6157
	DT_NO (combined)	(53+7)	0.5702	0.6941	0.4220	0.6001

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